

CAREER: HiGFM: High-Performance Graph Foundation Models through Algorithm and System Co-Design

Overview:

Graph-structured data appear across science and engineering, from program dependence graphs to scientific provenance graphs. Yet today’s graph neural networks (GNNs) must be trained from scratch for each new task and domain. As a result, they are data-hungry, fragile under distribution shift, and costly to deploy at scale. Graph Foundation Models (GFM) offer a better path. Much like BERT reshaped natural language processing, a GFM is pre-trained across diverse graphs and then adapted to downstream tasks without starting over. Realizing this vision, however, faces two intertwined obstacles. First, graphs lack universal token representations, and pre-training objectives that transfer across domains remain open problems. Second, existing HPC systems target dense tensor workloads and fit poorly with the sparse, irregular computation that graphs demand. To close both gaps, we propose HiGFM, a framework built on algorithm and system co-design. Thrust 1 develops scalable pre-training methods for GFMs. Thrust 2 builds HPC systems that make these methods practical at billion-node scale. Thrust 3 validates HiGFM on real applications and releases a community benchmark. On the educational side, we will introduce a new graduate course on Graph AI at UTA and expand undergraduate research for students from underrepresented groups.

Intellectual Merit:

HiGFM contributes across graph pre-training methodology and high-performance computing systems. First, in Thrust 1, we design new graph tokenization strategies and self-supervised pre-training objectives that capture transferable structural knowledge across graph families. Building on this, we establish formal bounds on the transfer gap as a function of structural distance between domains. This provides the theoretical grounding the field currently lacks. Second, in Thrust 2, we confront the graph-HPC impedance mismatch through structure-aware partitioning, asynchronous neighborhood sampling with computation and communication overlap, and graph-aware parameter-efficient fine-tuning. We also design an incremental inference runtime for dynamic graphs that avoids costly full re-embedding. All large-scale experiments will run on ORNL’s Frontier supercomputer in collaboration with Oak Ridge National Laboratory. Third, in Thrust 3, we validate HiGFM on two structurally distinct domains, program analysis for vulnerability detection and scientific data provenance for integrity verification. We then release a public benchmark suite that establishes a reproducible evaluation standard for the GFM community. Our preliminary work positions us uniquely to lead this agenda. This includes the Graph Intermediate Representation (GIR) for structural decomposition, the KSPine system for structure-aware shortest paths, and published GNN-based vulnerability detection results.

Broader Impacts:

Beyond research advances, HiGFM directly supports the national priority of AI for scientific discovery. First, our open-source benchmark suite will lower the barrier for researchers to evaluate and compare GFM progress in a reproducible way. Second, a new graduate course titled Graph AI Foundations and Systems will prepare the next generation of researchers at the intersection of graph algorithms and foundation models, with all materials released publicly to extend impact beyond UTA. Third, to broaden participation, we will integrate undergraduate students into Thrust 3 through structured research experiences, with priority recruitment through UTA’s LSAMP program at this Hispanic-Serving Institution. In addition, PhD students will gain hands-on HPC experience through the ORNL collaboration, preparing them for careers in academia, national laboratories, and industry. Finally, we will disseminate results through publications at top venues in machine learning, systems, and security, and through co-organization of a Graph AI community workshop.